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NON-SHOCK INITIATION MODEL FOR EXPLOSIVE FAMILIES: NUMERICAL RESULTS

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Abstract. A damage initiated reaction (DMGIR) computational model is being developed for the CTH shock physics code to predict the response of an explosive to non-shock mechanical insults. The distinguishing feature of this model is the introduction of a damage variable, which relates the evolution of damage to the initiation of reaction in the explosive, and its growth to detonation. The DMGIR model is a complement to the History Variable Reactive Burn (HVRB) model embedded in the current CTH code. Specifically designed experiments are supporting the development, implementation, and validation of the DMGIR numerical approach. PBXN-5 was the initial explosive material used experimentally to develop the DMGIR model. This explosive represents a family of plastically bonded explosives with good mechanical strength and rigid body properties. The model has been extended to cast explosives represented by Composition B.

Keywords: initiation, model, simulation, damage

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INTRODUCTION

Predicting the response of an explosive material subjected to a non-shock insult represents a notable change to the current initiation and growth models presently used. Generally, these models address a shock-to-detonation transition (SDT). Most SDT models use the pressure profile raised to an exponential value multiplied by the pulse duration as their critical energy initiation criteria as shown

$$E_c \propto P^n \tau_c \quad (1)$$

where E_c is the critical energy required to generate an initiation, P is the amplitude of the shock pressure, n is an exponential value of $1 < n < 2$, and τ_c is the pressure pulse duration. The DMGIR model introduces damage variable, which relates the evolution of damage to the initiation of a reaction and subsequent growth to a detonation [1].

NUMERICAL APPROACH

The DMGIR model was embedded into CTH to use the well-validated capabilities of the CTH code. Comparisons of DMGIR model predictions with recent experiments are presented for non-shock initiation in the plastically bonded explosive PBXN-5 and cast Composition B. Future work is intended to expand the DMGIR model for moldable plastically bonded explosives.

HISTORY VARIABLE REACTIVE BURN (HVRB) MODEL

The HVRB is a pressure-based model for a shock-induced initiation in a heterogeneous explosive material [2]. Equation (2) shows the pressure relation used to generate an initiation

$$\phi(t) = \frac{1}{\tau_0} \int_0^t \left[\frac{(P - P_I)}{P_R} \right]^z dt \quad (2)$$

where ϕ is the history variable, τ_0 is a normalization constant of 1 μ s, P is pressure, P_I is the threshold pressure, P_R is the scaling pressure, and Z is the pressure exponent. The history variable is calibrated by fitting the parameters P_I , P_R , and Z to Pop-plot data. Most pressure-based reaction models use the pressure-dependent rate law [2],

$$\frac{d\lambda}{dt} = A(1-\lambda)^m f(P) \quad (3)$$

which sufficiently describes the shock initiation of heterogeneous explosives. The HRVB model uses a simplified version of equation (3),

$$\begin{aligned} \lambda(t) &= \min\left(1, \phi(t)^M\right) \\ &= 1 - \left[1 - \frac{\phi(t)^M}{X}\right]^X \end{aligned} \quad (4)$$

where λ is the extent of reaction variable, M controls the time delay to pressure buildup behind the shock wave, X governs the rate of reaction. The history variable, ϕ , in equation (4) is dimensionless, and through its integration gives the pressure-time history imposed upon an explosive sample.

The HVRB model uses a mixture rule [2] to yield the following composite equations of state,

$$P(\rho, T, \lambda) = P_U(\rho, T)(1-\lambda) + P_R(\rho, T)\lambda \quad (5)$$

$$E(\rho, T, \lambda) = E_U(\rho, T)(1-\lambda) + E_R(\rho, T)\lambda \quad (6)$$

to obtain the pressure and energy, where the subscript U identifies the equation of state for the unreacted (solid state) portion of the material and the subscript R identifies the equation of state for the reacted (gaseous state or detonation products) portion of the material. When the HVRB is used, the equations of state for the unreacted and reacted phases are usually the Mie-Grüneisen and JWL equations of state, respectively. Once $\lambda = 1$, the mixture equation of state (5, 6) reduces to the reacted-phase equation of state.

DAMAGE-INITIATED REACTION (DMGIR) MODEL

The DMGIR modifies the HVRB by replacing the history variable with a damage variable, producing a new extent-of-reaction equation

$$\begin{aligned} \lambda_{DMGIR}(t) &= \min\left(1, D_{DMGIR}^M\right) \\ &= 1 - \left[1 - \frac{D_{DMGIR}^M}{X}\right]^X \end{aligned} \quad (7)$$

The damage variable (derived from the Johnson-Cook damage model) is

$$D_{DMGIR} = \int \frac{d\varepsilon}{\varepsilon_R} \quad (8)$$

In equation (8), $d\varepsilon$ is the incremental equivalent plastic strain. The reference strain ε_R is

$$\begin{aligned} \varepsilon_R &= \underbrace{\left[D_{R1} + D_{R2}e^{-D_{R3}\sigma^*}\right]}_{\text{maximum threshold}} \times \\ &\quad \underbrace{\left[1 - D_{R4} \ln \varepsilon^*\right]}_{\text{strain-rate}} \times \underbrace{\left[1 - D_{R5}T_R^{*m}\right]}_{\text{temperature}} \end{aligned} \quad (9)$$

where D_{R1} , D_{R2} , D_{R3} , D_{R4} , and D_{R5} are material constants and

$$\varepsilon^* = \frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \quad (\dot{\varepsilon}_0 = 1.0 \text{ s}^{-1})$$

$$T_R^* = \frac{(T - T_{room})}{(T_{melt} - T_{room})} \text{ for } 0 \leq T^* \leq 1.0$$

$$\sigma^* = \frac{\sigma_m}{\sigma_{eff}} = \frac{\text{mean stress}}{\text{effective stress}}$$

Table 1 shows the current experimentally derived coefficients for each explosive material tested.

Table 1 DMGIR coefficients

Explosive material	D_{R1}	D_{R2}	D_{R3}	D_{R4}	D_{R5}
PBXN-5	5.0	0.0	-1.0	-0.0592	0.0
Comp B	3.0	0.0	-1.0	-0.0358	0.0

The DMGIR relates damage growth to reaction initiation and detonation. The D_{R1} coefficient sets a maximum threshold value, while D_{R4} increases the sensitivity of the explosive material by reducing the overall reference strain needed to generate a reaction.

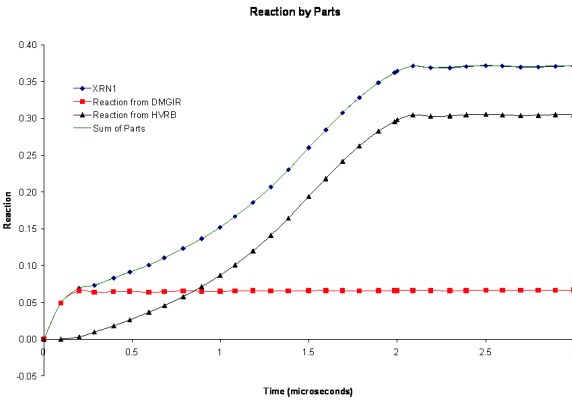


Figure 1 Reaction by parts.

During numerical calculations, the extent of reaction uses the higher value of D or ϕ for each cell/time step to calculate the fraction and rate of reaction. In Figure 1, the DMGIR dominates the extent of reaction during the early stages. At later time, the pressure and duration are enough to allow the HVRB model to dominate the reaction process.

RESULTS

Since the DMGIR depends on the evolution of damage, it is essential to develop and implement material models that accurately depict the behavior of the reactive material under dynamic loading. The constitutive models used for each explosive material are the Johnson-Cook strength and Johnson-Cook damage or fracture model [3].

$$Y = (A + B\varepsilon^n)(1 + C \ln \dot{\varepsilon})(1 - T^{*m})$$

$$D = \int \frac{\varepsilon_{ps}}{\varepsilon_f} \quad (10)$$

$$\varepsilon_f = (D_1 + D_2 e^{-D_3 \sigma^*}) \times (1 + D_4 \ln(\dot{\varepsilon})) \times (1 + D_5 T^*)$$

where A , B , C , n , m , D_1 , D_2 , D_3 , D_4 , and D_5 are material constants. Based on quasi-static and dynamic strain rate data obtained by Los Alamos National Laboratories [4], Table 2 and Table 3 show the derived Johnson-Cook strength and damage coefficient for PBXN-5 and Comp B.

Table 2 Johnson-Cook strength coefficients

Explosive material	A (MPa)	B (MPa)	C	m	n
PBXN-5	0.8667	445	1.3766	0.78125	0.8726
Comp B	0.42	237	0.498	1.767	0.73

Table 3 Johnson-Cook damage coefficients

Explosive material	D_1	D_2	D_3	D_4	D_5
PBXN-5	0.03574	0	1	-0.104	-0.54
Comp B	0.42	237	0.498	1.767	0.73

With the newly develop strength and damage material models for PBXN-5 and Comp B in CTH, the DMGIR agrees with experimental data as shown in Figure 2 through Figure 4.

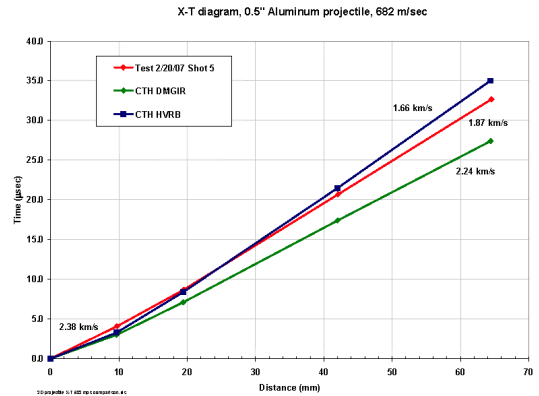


Figure 2 Aluminum projectile impact, PBXN-5 billet at 682 m/s, no detonation.

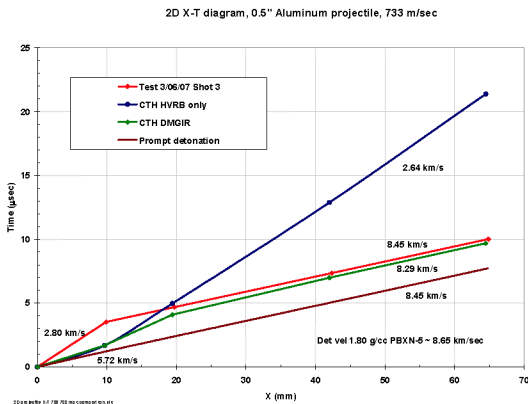


Figure 3 Aluminum projectile impact, PBXN-5 billet at 733 m/s, detonation.

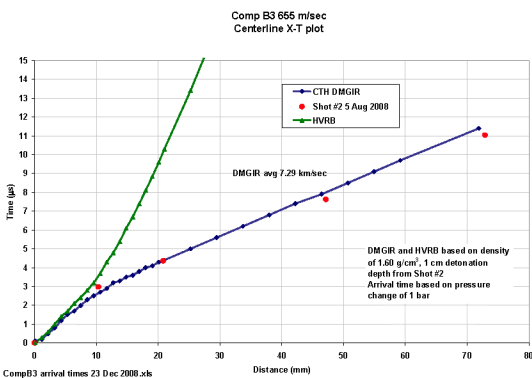


Figure 4 Brass projectile impact, Comp B billet at 655 m/s, detonation.

References 1 and 6 provide a more thorough explanation of the numerical approach with further comparisons. References 1 and 5 describe the experimental testing program and results.

SUMMARY

During damage evolution in a reactive material, the induced damage generates new surfaces and void space. Crush-up of the void space and related shear dominate the early stages of the initiation process. The DMGIR model captures the initial portion of the process.

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